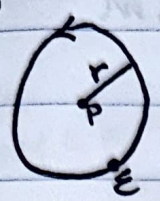
HW #8

1)



Given: $r = 5.30 \times 10^{-11}$ m

$s = 2.20 \times 10^{6}$ m/s

$m = 9.11 \times 10^{-31}$ kg

2) $C = 2\pi r$

$= 2 \cdot 5.30 \times 10^{-11}\,m \cdot \pi$

$= 3.33 \times 10^{-10}$ m

$T = C/s$

$= 1.51 \times 10^{-16}$ s

b) $\cancel{\vec{F}_{net} = m\vec{a}}$

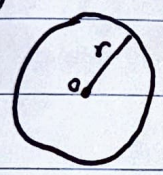
$\cancel{= m}$

$F_{net} = ma$

$= m\frac{s^2}{r}$

$= 8.32 \times 10^{-30}$ kg m/s²

2)



Given: $\mu_s = .6$

$r = 30.5$ m

$V = \frac{2\pi r}{T}$

$a = \frac{V}{T}$

$= \frac{2\pi r}{T^2}$

$F = ma$

$= m\frac{2\pi r}{T^2}$

$\mu_{s,max} = \mu_s F$

$= \mu_s \frac{2\pi m r}{T^2}$

$a_{,max} = \mu_s \frac{2\pi m r}{T^2} \div m$

$= \mu_s \frac{2\pi r}{T^2}$

$V_{,max} = \mu_s \frac{2\pi r}{T^2} \div T$

$= \mu_s \frac{2\pi r}{T}$

$\cancel{V = .6 \cdot 2\pi}$

$V = \frac{2\pi r}{T}$

$a = \frac{V^2}{r}$

$= \left(\frac{2\pi r}{T}\right)^2 \frac{1}{r}$

$F = ma = mr\left(\frac{2\pi}{T}\right)^2$

$f_{s,max} = \mu_s F$

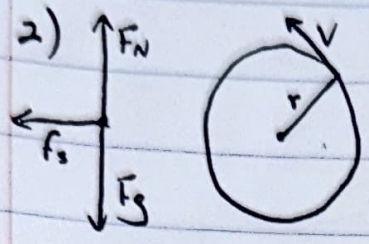
$= \mu_s mr\frac{4\pi^2}{T^2}$

$a_{,max} = f_{s,max} / m$

$v_{,max} = f_{s,max} / mT$

$=$

2)



Given: $\mu_s = .6$, $r = 30.6\,m$

$$\vec{F}_{net} = \vec{F}_N + \vec{F}_g + \vec{f}_s$$
$$= \vec{f}_s$$

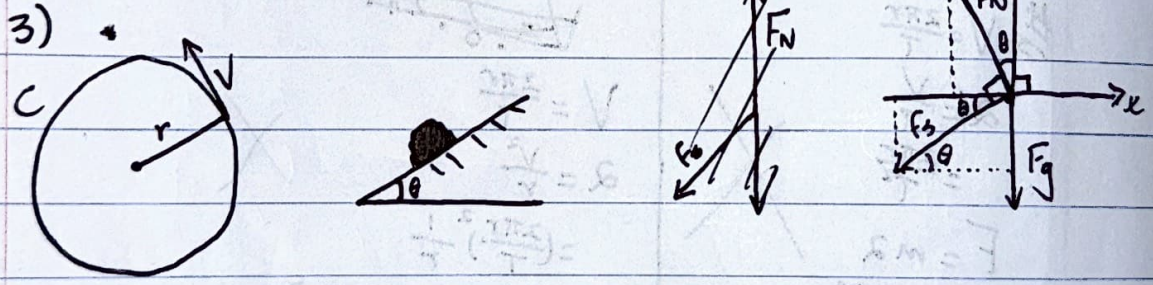
$$V = \frac{2\pi r}{T}$$
$$a = \frac{V^2}{r}$$

$f_{s,max}$

$$\cancel{f_s} = \mu_s F_N = -F_g \mu_s$$
$$= -\mu_s mg$$

$$V_{,max} = f_{s,max}/m \cdot t$$
$$= -\mu_s g \cdot t$$
$$= \cancel{\mu_s} -.6 \cdot -9.81\,m/s^2 \cdot s$$
$$= 5.886\,m/s$$

3)



$$\vec{F}_g = mg$$
$$\cancel{\vec{F}_N = -mg\cos\theta}$$
$$\cancel{F_{Nx} = -mg\cos\theta}$$
$$\cancel{F_{Ny} = -mg\sin\theta}$$

$$f_{s,max} = \mu_s F_N$$
$$\cancel{f_{sx} = \mu_s F_{Nx}}$$
$$\cancel{f_{sy} = \mu_s F_{Ny}}$$
$$f_{sx,max} = \mu_s F_{Nx}$$
$$f_{sy,max} = \mu_s F_{Ny}$$

$$C = 2\pi r$$
$$T = \frac{2\pi r}{V}$$
$$\vec{a} = \frac{4\pi^2 r}{T^2}$$
$$\vec{F}_{net} = \vec{F}_g + \vec{F}_N + \vec{f}_s$$
$$= m\frac{4\pi^2 r}{T^2}$$

Given: $r = 80.0\,m$, $\theta = 15^\circ$, $V = 30.0\,m/s$

$$\vec{F}_{net} = m\frac{4\pi^2 r}{T^2}$$
$$= m\frac{4\pi^2 r}{1}\cdot\frac{V^2}{4\pi^2 r^2}$$
$$= \frac{mV^2}{r}$$

$$F_{Ny} = F_g + f_{sy}$$
$$F_N\cos\theta = mg + \mu F_N\sin\theta$$
$$F_N = \frac{mg}{\cos\theta - \mu\sin\theta}$$

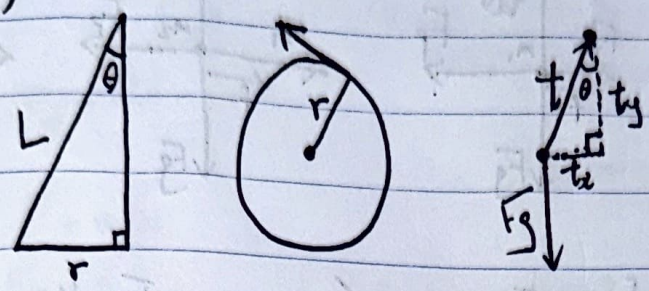
$$F_N\sin\theta + f_{sx} = \frac{mV^2}{r}$$
$$F_N\sin\theta + \mu F_N\cos\theta = \frac{mV^2}{r}$$
$$F_N(\sin\theta + \mu\cos\theta) = \frac{mV^2}{r}$$
$$\frac{g(\sin\theta + \mu\cos\theta)}{\cos\theta - \mu\sin\theta} = \frac{V^2}{r}$$

$$\cancel{\vec{F}_{net} = \vec{F}_g + \vec{F}_N + \vec{f}_s}$$
$$\cancel{= (0\hat{i} + m9.81\hat{j}) + (-m\cancel{9.81}\hat{i} + m9.81\hat{j}) + f_s}$$

$$g(\sin\theta + \mu\cos\theta) = \frac{V^2}{r}(\cos\theta - \mu\sin\theta)$$
$$g\sin\theta + \mu g\cos\theta = \frac{V^2\cos\theta}{r} - \frac{\mu V^2\sin\theta}{r}$$
$$\mu\left(g\cos\theta + \frac{V^2}{r}\sin\theta\right) = \frac{V^2\cos\theta}{r} - g\sin\theta$$

$$\mu = \left(\frac{V^2\cos\theta}{r} - g\sin\theta\right) / \left(g\cos\theta + \frac{V^2}{r}\sin\theta\right)$$
$$\cancel{= \left(\frac{(30.0\,m/s)^2\cdot\cos 15^\circ}{80\,m} - 9.81\,m/s^2\cdot\sin 15^\circ\right) / \left(-9.81\,m/s^2\cdot\cos 15^\circ + \frac{(30\,m/s)^2\cdot\sin}{80\,m}\right.}$$
$$\cancel{= (10.86\,m/s^2 + 2.54\,m/s^2) / (-9.47\,m/s^2 + 2.91\,m/s^2)}$$
$$\cancel{= 13.40\,m/s^2 / -6.558\,m/s^2}$$
$$= \left(\frac{(30\,m/s)^2\cdot\cos 15^\circ}{80\,m} - 9.81\,m/s^2\cdot\sin 15^\circ\right) / \left(9.81\,m/s^2\cdot\cos 15^\circ + \frac{(30\,m/s)^2\cdot\sin 15^\circ}{80\,m}\right)$$
$$= (10.86\,m/s^2 - 2.54\,m/s^2) / (9.47\,m/s^2 + 2.91\,m/s^2)$$
$$= \cancel{13.40}\,8.32\,m/s^2 / 12.38\,m/s^2$$
$$= .67$$

4)



$\vec{F}_{net} = m\frac{4\pi^2 r}{T^2}$

$+F_g + t_y = 0 = F_{net\,y}$

$0 + t_x = t_x = F_{net\,x}$

$t_y = -F_g$

$= mg$

$t_x = t\sin\theta$

$t = \frac{t_y}{\cos\theta}$

$= \frac{mg}{\cos\theta}$

$4m\frac{\pi^2 r}{T^2} = t\sin\theta$

$r = L\sin\theta$

$4m\frac{\pi^2 r}{T^2} = \frac{mg}{\cos\theta}\sin\theta$

$4m\frac{\pi^2 r}{T^2} = mg\tan\theta$

$4\pi^2 r\cdot\frac{1}{T^2} = g\tan\theta$

$T^2 = 4\frac{\pi^2 r}{g\tan\theta}$

$= 4\frac{\pi^2 L\sin\theta}{g\tan\theta}$

$T = \sqrt{4\frac{\pi^2 L\sin\theta}{g\tan\theta}}$

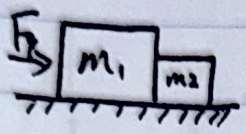
$T = 2\pi\sqrt{\frac{L\sin\theta}{g\tan\theta}}$

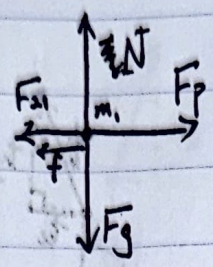
$= 2\pi\sqrt{\frac{L}{g}\sin\theta\cdot\frac{1}{\tan\theta}}$

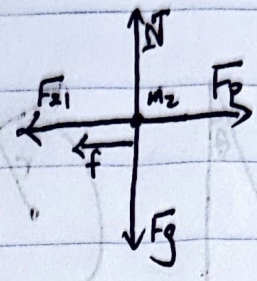
$= 2\pi\sqrt{\frac{L}{g}\sin\theta\cdot\frac{\cos\theta}{\sin\theta}}$

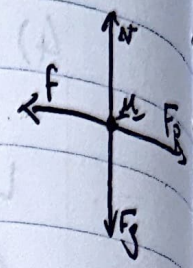
$= 2\pi\sqrt{\frac{L\cos\theta}{g}}$

5)









$M = m_1 + m_2$

$\vec{N} = \vec{F_g}$

$\vec{F}_{net_y} = \vec{N} - \vec{F_g} = 0$

$\vec{F}_{net_x} = \vec{F_p} + f$

$= \vec{F_p} + \mu_k N$

$= \vec{F_p} + \mu_k M g$

$\vec{F}_{net} = \vec{F_p} + \mu_k M g$

a) $\vec{a} = \dfrac{\vec{F_p} + \mu_k M g}{M}$

$= \dfrac{20N + .2 \cdot 8kg \cdot 9.81 m/s^2}{8kg}$

$= 4.30N/8kg$

$= .54\ m/s^2$

$f = \mu_k N$

$= \mu_k M g$

Given: $m_1 = 5.00kg$, $m_2 = 3.00kg$, $\mu_k = .200$, $F_p = 20N$

$\vec{F}_{net_{m_1}} = \vec{F_p} + \mu_k m_1 g$

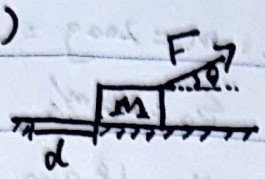
b) $\vec{F}_{net_{m_1}} = \vec{F}_{net} + \vec{F}_{21}$

$\vec{F}_{21} = (\vec{F_p} + \mu_k M g) - (\vec{F_p} + \mu_k m_1 g)$

$= \mu_k g (M - m_1)$

$= -5.886N$

6)



Given: $F = 10.0N$, $f = 0$;
$\theta = 37°$, $d = 3.00\,m$,
$m = 2.00\,kg$, $F_x = 10N\cos 37°$

$F = m\frac{d}{t^2}$

$\frac{F}{m} = \frac{d}{t^2}$

$\frac{F}{m}t^2 = d$

$t^2 = \frac{dm}{F}$

$t = \sqrt{\frac{dm}{F}}$

$= \sqrt{\frac{3m \cdot 2kg}{10N}}$

$= \sqrt{\frac{6mkg}{10kg\frac{m}{s^2}}}$

$= \sqrt{.6s^2}$

$= .77s$

$K_{total} = \frac{1}{2}mV^2$

$= \frac{1}{2}m\frac{d^2}{t^2}$

$= \frac{1}{2} \cdot 2.00kg \cdot \frac{3.00m^2}{.77s^2}$

$= 3.90J$

X

$F = m\frac{d}{t^2}$

$t = \sqrt{\frac{dm}{F}}$

$= \sqrt{\frac{3m \cdot 2kg}{10N\cos 37°}}$

$= \sqrt{\frac{6mkg}{7.99kg\frac{m}{s^2}}}$

$= \sqrt{.75s^2}$

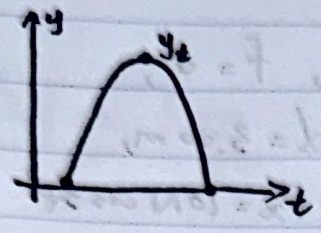
$= .87s$

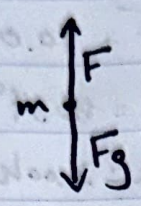
$K = \frac{1}{2}mV^2 = \frac{1}{2}m\frac{d^2}{t^2}$

$= \frac{1}{2} \cdot 2kg \cdot \frac{3.00m^2}{.87s^2}$

$= 11.89J$

7)





Given: $m = 200g = .200kg$,
$S_0 = 20.0 m/s$,
$y_t = 18.0m$

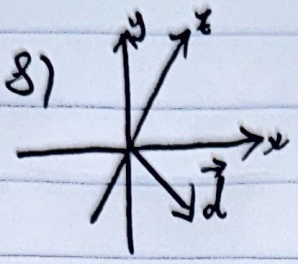
a) $\Delta K = K_f - K_0$
$= \frac{1}{2}mV_f^2 - \frac{1}{2}mV_0^2$
$= \frac{1}{2}m(V_f^2 - V_0^2)$
$= \frac{1}{2} \cdot .2kg \cdot (0^2 - 20 m/s^2)$
$= .1kg \cdot -400 \frac{m^2}{s^2}$
$= -40J$

b) $W = F_g y_t \cos\theta$
$= mg y_t \cos\theta$
$= .2kg \cdot -9.81 m/s^2 \cdot 18 m \cdot \cos 180°$
$= 35.3J$

c) in (a), we figured out the Total Change of Kinetic Energy. which is $-40J$.
while in (b), what we found is just the Change of Kinetic Energy due to Gravity, which $35.3J$.

though the two concepts are very similar, but still refer to the different stuff.

8)



Given: $F = 12.0N$,

$\vec{d} = (2.00\hat{i} + -4.00\hat{j} + 3.00\hat{k})$

$d = \sqrt{4+16+9}$

$= \sqrt{29} = 5.39\ m$

$W = F \cdot d \cos\theta$

$\cos\theta = \frac{W}{Fd}$

$\theta = \cos^{-1}\left(\frac{W}{Fd}\right)$

a) $\theta = \cos^{-1}\left(\frac{30J}{12N \cdot 5.39m}\right)$

$= \cos^{-1}(.46)$

$= 62.37°$

b) $\theta = \cos^{-1}\left(\frac{-30J}{12N \cdot 5.39m}\right)$

$= \cos^{-1}(-.46)$

$= 117.63°$